Huawei Cloud EI – API Services

Foreword

- This chapter describes the functions and application scenarios of Huawei Cloud EI API services, including Image Recognition, ImageSearch, Content Moderation, Optical Character Recognition (OCR), and Speech Interaction Service (SIS). It also provides an example of how to use these services.

Objectives

- Upon completion of this course, you will understand:
  - Main functions of API services
  - Application scenarios of API services
  - How to use API services

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Contents

1. **Image Recognition**
2. ImageSearch
3. Content Moderation
4. OCR
5. SIS
6. Invoking API Services

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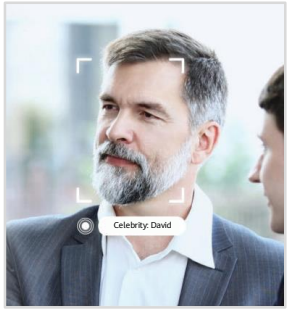
What Is Image Recognition?

- Image Recognition is a technology that uses computers to analyze and understand images to identify objects in different modes. Image Recognition provides six functions to enable tagging for all media and image optimization.

Animal and plant recognition

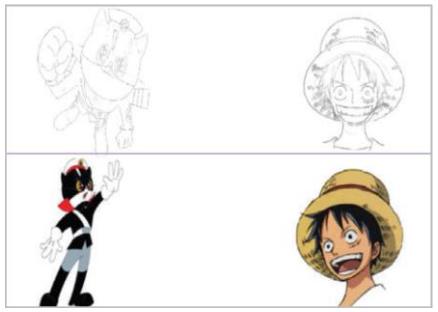


Celebrity recognition

Home recognition

Face cutout

Animation style generation



Animation coloring

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Application Scenario – Scenario Analysis

- Image Tagging can accurately recognize video and image content, thereby improving retrieval efficiency and precision. It makes personalized recommendation and content retrieval and distribution more effective.

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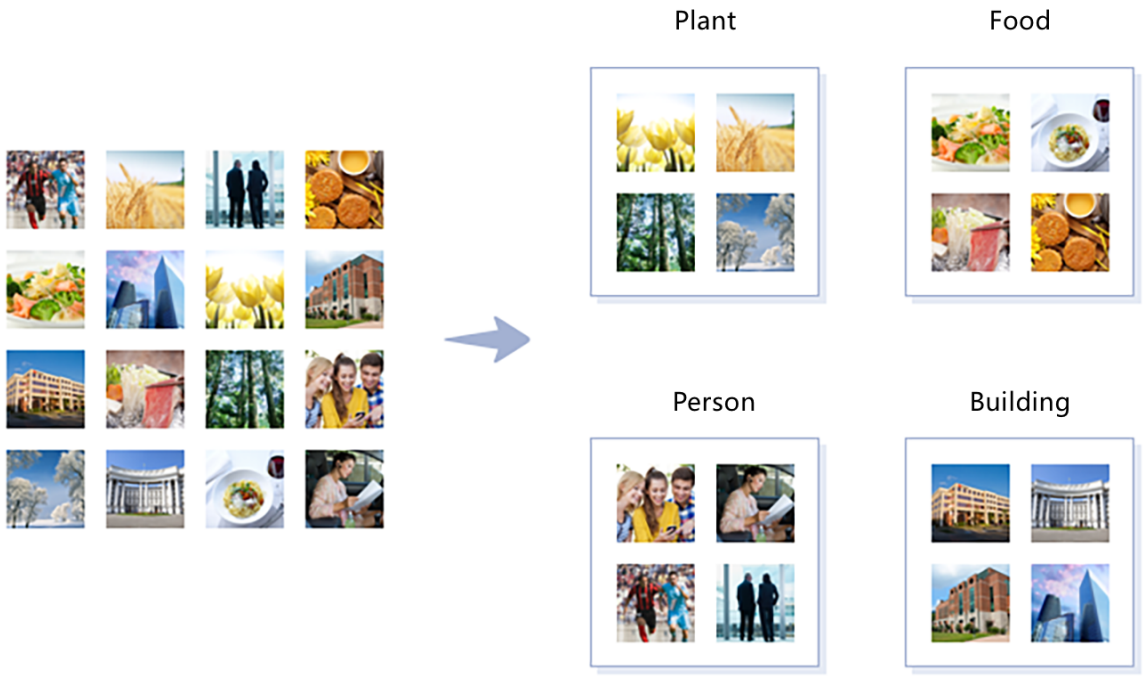
Application Scenario – Smart Albums

- Image Recognition can recognize approximately 10,000 tags, such as plant, food, and work, giving you a better experience in managing albums by tag.



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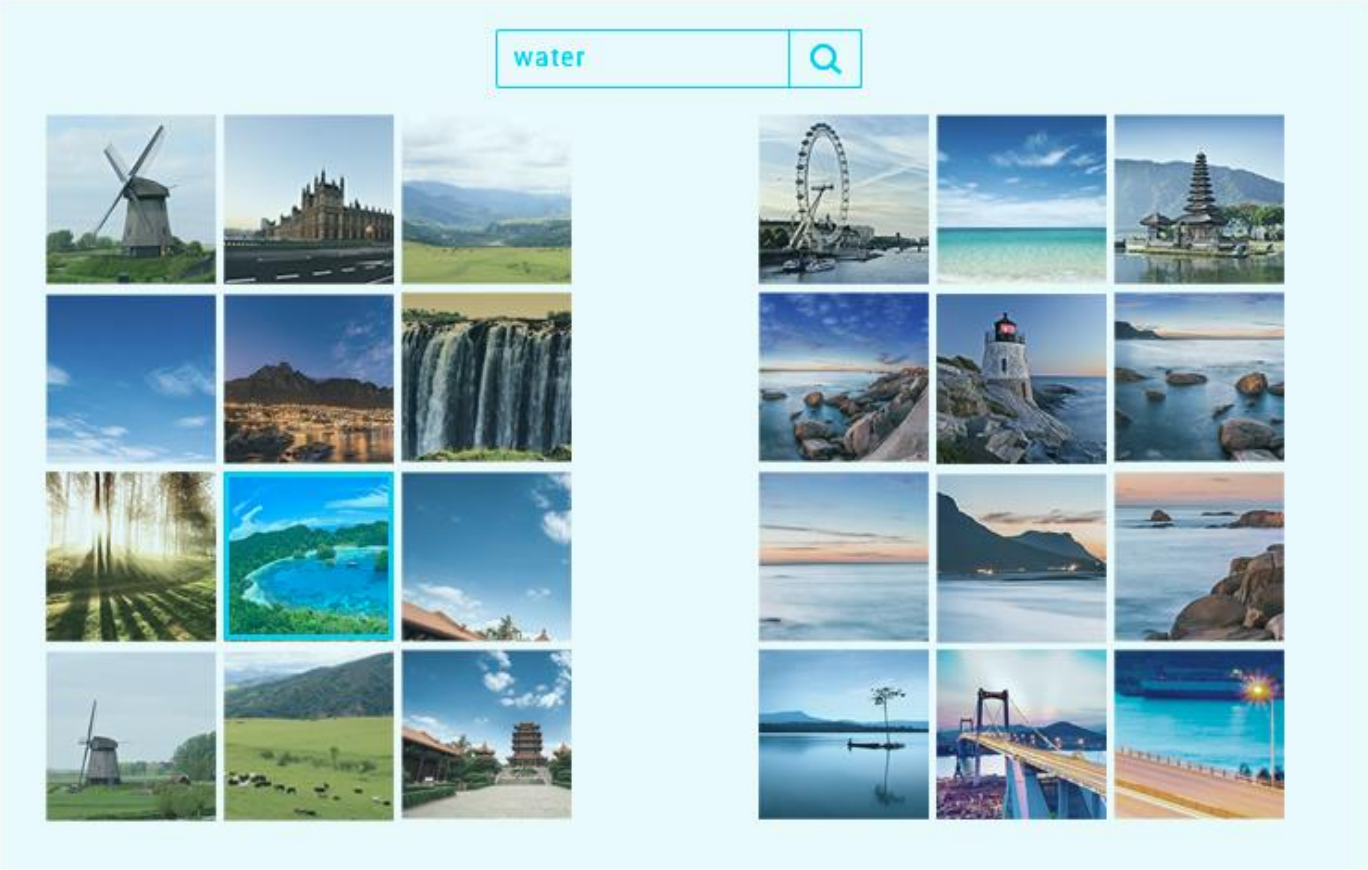
Application Scenario – Object Detection

- At construction sites, a customized object detection system can monitor whether onsite personnel wear safety helmets in real time, which helps reduce security risks.

Application Scenario – Image Search

- The image search technology helps you quickly find the images that match the keywords or images you input.



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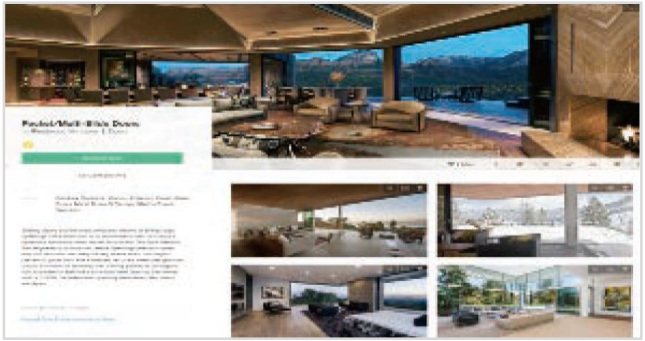
What Is ImageSearch?

- ImageSearch leverages cutting-edge deep learning and image recognition technologies to deliver service- and industry-specific feature vectorization and search capabilities that help you search images from a specified gallery using the exact or fuzzy match option.

Similar images search

Industrial parts search

Unattended supermarkets



Architectural design search

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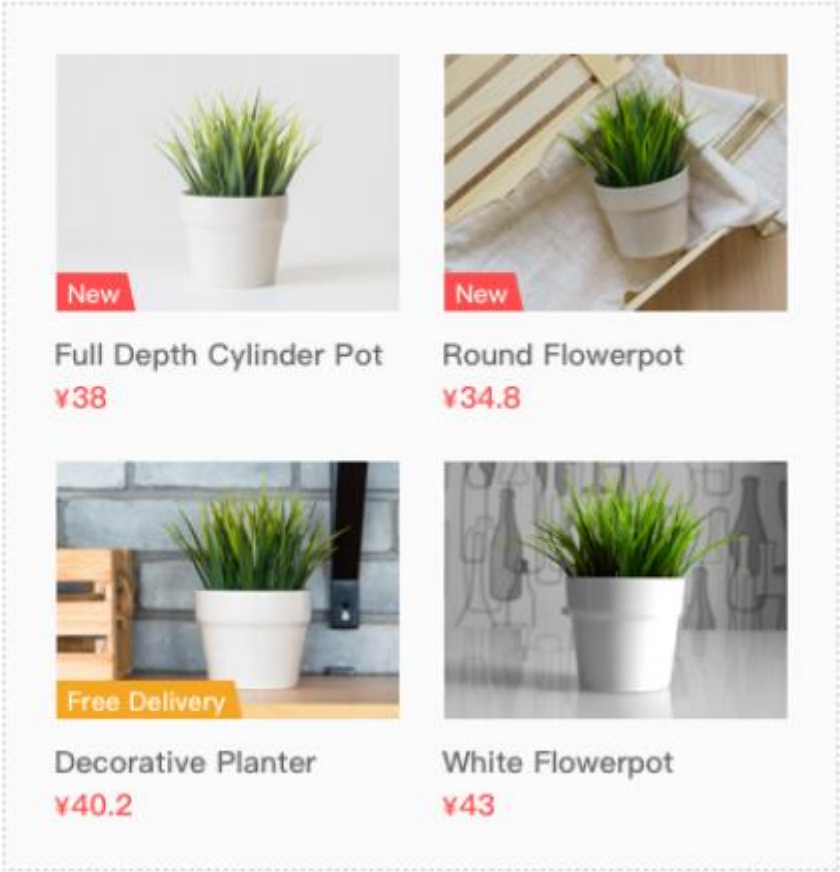
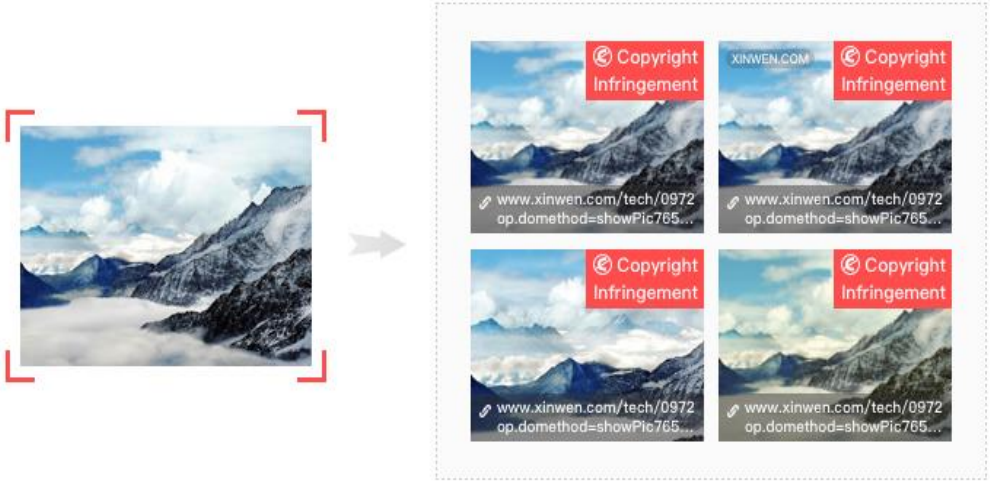
Application Scenario – Product Image Search

- Product image search
  - ImageSearch allows you to search for a product or similar products by image. You can search for an image from hundreds of millions of images in seconds and update data in real time. ImageSearch simplifies product search, allowing you to search for a product without providing its details, helping increase sales while improving UX.



Application Scenario – Product Recommendation & Copyrighted Image Search

- Product recommendation
  - ImageSearch allows users to search for a product or similar products by image, and recommends related products.
- Copyrighted image search
  - ImageSearch can quickly locate infringed images in galleries. This helps website galleries protect their rights and interests.



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What Is Content Moderation? (1)

- Content Moderation uses cutting-edge image, text, and video recognition techniques that precisely detect advertisements, pornographic or terrorism-related content, and politically sensitive information, reducing any compliance risks for your business and improving review efficiency.

Moderation scenarios

Video | Image | Text

Compliance review

Pornography detection:
Identifies pornographic content in images and delivers three confidence scores indicating pornographic, sexy, and normal images, respectively.

Terrorism detection:
Checks whether images contain guns, bloody or horrible content, or cult information.

Image text moderation:
Checks whether images contain QR codes or non-compliant text.

Political figure detection:
Checks whether images contain politically sensitive figures or misbehaving celebrities.

Inappropriate content detection:
Checks whether images contain inappropriate content such as nausea, smoking, and gambling.

Text moderation:
Checks whether text contains pornographic content, political content, terrorism-related content, advertisements, abuse, contraband, or spams.

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What Is Content Moderation? (2)

Content Review
Detects and flags non-compliant content for quick management, lowering legal exposure.

Live Video
Monitors, identifies, and flags live video channels with offensive content in real time.

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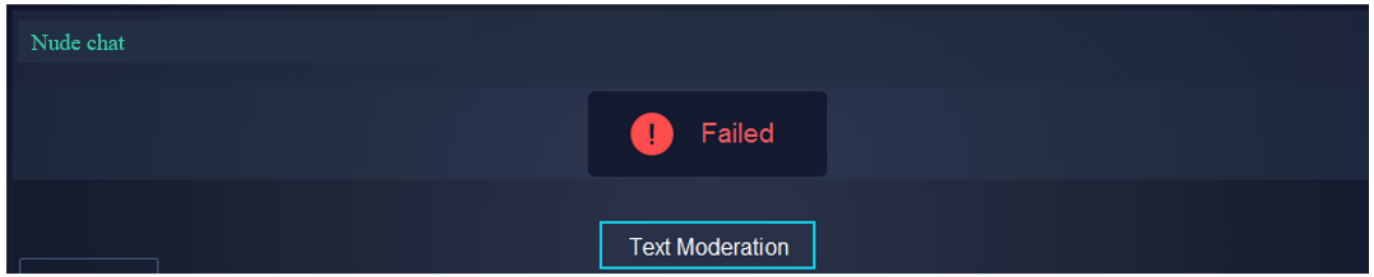
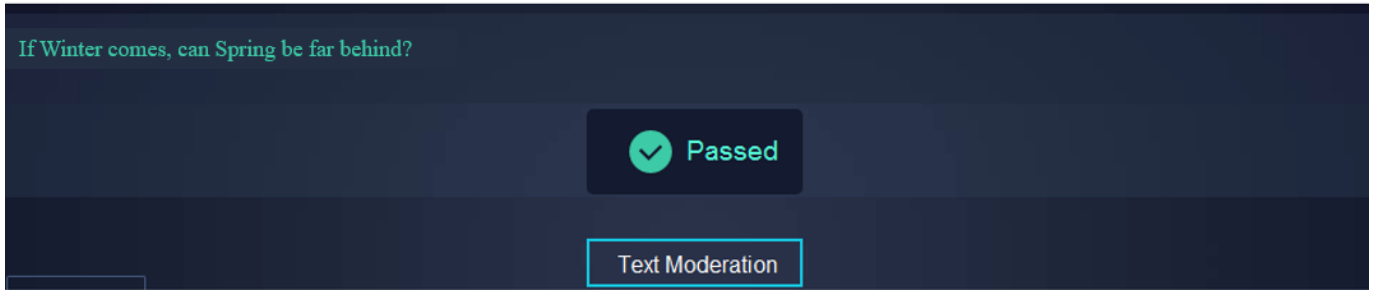
Text, Audio, and Video Moderation

- Text moderation uses a large word library, deep learning, graph convolutional network (GCN), and text classification models to automatically identify political and pornographic content, advertisements, abuse, and spams in text. It connects to the Cyberspace Administration of China (CAC) around the clock and can be used to review comments, SMS messages, nicknames, bullet screens, and signatures.
- Audio moderation uses deep learning-based speech recognition technologies and text moderation to filter out malicious information in a speech, such as asthma, terrorism-related and political content, advertisements, and abuse. It can be used to review audio livestreaming, radio, and music.
- Video moderation reviews the images, text, and audio of offline videos and video livestreams.





Application Scenarios of Content Moderation

Short videos, online shopping, forums, chat, livestreaming, advertisements, and media

Image moderation

- Live video
- Online shopping
- Internet forum

Text moderation

- E-commerce comment screening
- Media content review
- User nickname review
- Bullet comment review
- Chat content review

Video moderation

- Video platforms/ communities
- Video chat
- Online education

9 types

Pornographic, sexy, terrorism-related, and political content, spams, abuse, advertisements, contraband, and picture-in-picture

0.1s

Time required for processing an image using an AI chip

95%

Accuracy

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Application Scenarios of Content Moderation

Apps	**Huawei AppGallery** is the official app distribution platform for Huawei devices. It allows developers around the world to develop apps. These apps need to be moderated so that they abide by the laws, regulations, and religious customs of the countries or regions where they are available. **Huawei Cloud Content Moderation** provides all-round moderation policies to help Huawei AppGallery securely go global.
Media	**Media enterprises** want to detect pornography, advertisements, and terrorism-related and political content in images, text, videos, livestreams, comments, user profiles, and audio. **Huawei Cloud Content Moderation** provides a multimodal content moderation system which helps them effectively cleanse their products and improve user experience. The recall rate is 95% and accuracy is 97%.
E-commerce	**In shopping livestreams**, pornography, advertisements, and terrorism-related and political content in the bullet screens, images, and videos must be moderated. **Huawei Cloud Content Moderation** helps customers improve the moderation efficiency by over 90% and handle videos with violations in three minutes.
Gaming	**In game scenarios**, there is a large amount of data that needs moderation. The pornography, advertisements, and terrorism-related and political content in players' nicknames, avatars, signatures, chats, and chat backgrounds need to be moderated. **Huawei Cloud Content Moderation** makes games safer, improves user experience, and saves costs.
Cloud storage	**Cloud disks** store large amounts of text, images, and videos, which may be subject to security supervision if pornographic, terrorism-related, and political content is detected. **Huawei Cloud Content Moderation** provides a multimodal content moderation system and custom moderation policies which significantly save costs compared with manual moderation and effectively cleanse products.

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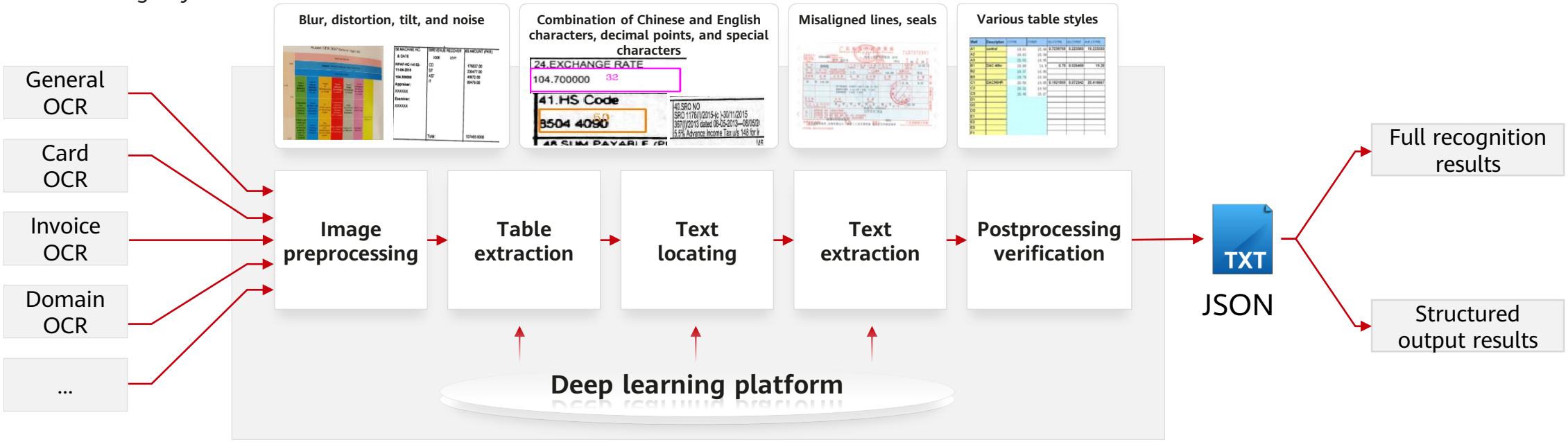
Introduction to OCR

Optical Character Recognition (OCR)

OCR technology automatically detects and converts text in images or scanned documents into editable text, eliminating the need for manual data entry. It has a wide range of applications, including card identification, license plate recognition, healthcare, PDF-to-Word conversion, photo and screenshot recognition, web image recognition, autonomous driving, paperless offices, document editing and proofreading, logistics sorting, public opinion monitoring, document retrieval, subtitle recognition, and literature search. OCR can recognize both printed and handwritten characters using a general process that involves identifying the character area, dividing it into individual characters, classifying and recognizing the characters, and post-processing and correcting any errors.



Training with large volumes of raw/analog data

http://3ms.huawei.com/km/blogs/details/7839677?l=en

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Functions of Huawei Cloud OCR

Full-text recognition

General OCR

- General Text
- General Table
- Web Image

Extensible

Structured product

Card OCR

- ID Card
- Driving/Vehicle License
- Passport
- Bank Card
- Business Card

Extensible

Domain OCR

- Electronic Waybill
- Financial Report

Extensible

1. OCR services developed in China can be exported and customized for use in other countries using customer-provided samples.
2. Huawei Cloud can also provide OCR software that supports local languages through partnerships with local companies.

- Myanmar ID Card
- Myanmar Driving License
- Thailand ID Card
- Passport
- India PAN Card/ID Card

Available

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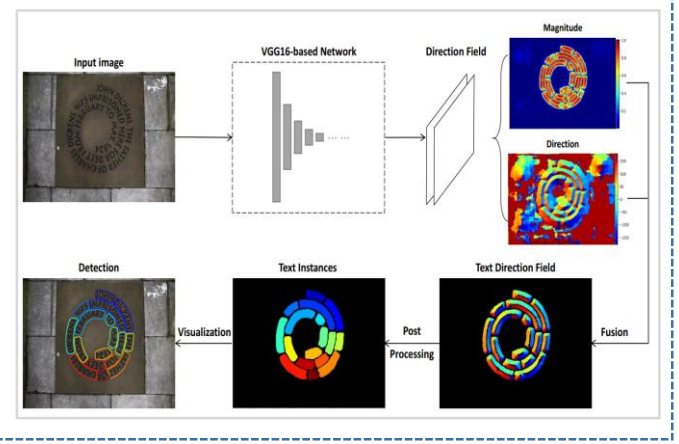
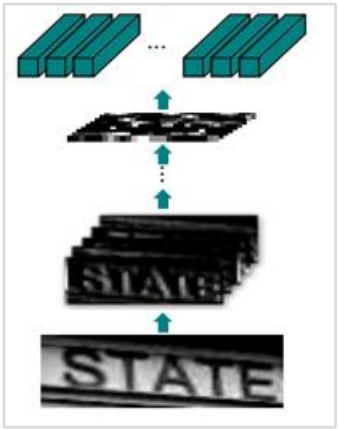
OCR Service Process

Structured information input

Structured result output

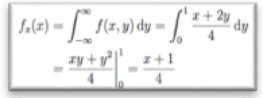


Character locating

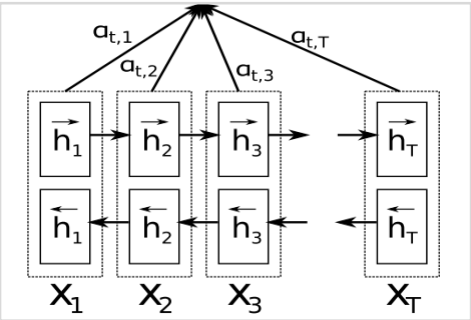


MapReduce Service

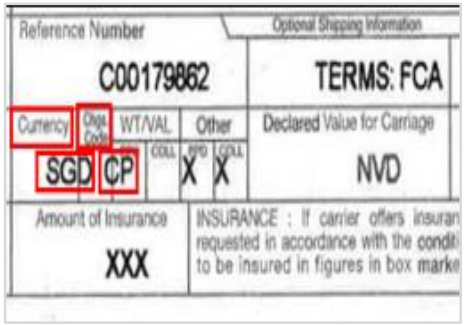
AI development platform ModelArts



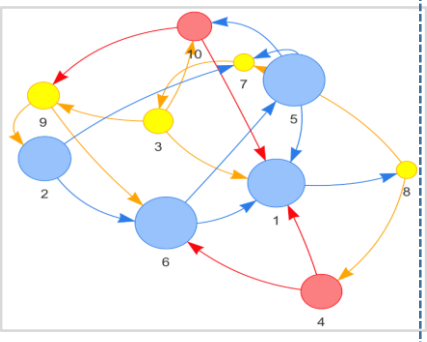
Character recognition



Natural language processing



Professional domain knowledge



Graph convolutional neural network — graph analysis

Structured information extraction

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What Is Structured Information Extraction

Text detection

Character recognition

Results

"Da Bingfeng"
"Name"
"Gender (Female)"
"Ethnicity (Mu Lao)"
"September 28, 2041"
"Date of Birth"
"Lian Feng Town, Liancheng County, Longyan City, Fujian Province"
"Address"
"No. 33 Lane, Lianhuatang Road, Lianhua Village"
"205046033595049978"
"ID No."

Address?

Text detection

Character recognition

Structured information extraction algorithm

Results	
Key	Value
Name	Da Bingfeng
Gender	Female
Ethnicity	Mu Lao
Date of Birth	September 28, 2041
Address	No. 33 Lane, Lianhuatang Road, Lianhua Village, Lian Feng Town, Liancheng County, Longyan City, Fujian Province
ID No.	205046033595049978

Structured information extraction:

The recognition results are returned as key-value pairs.

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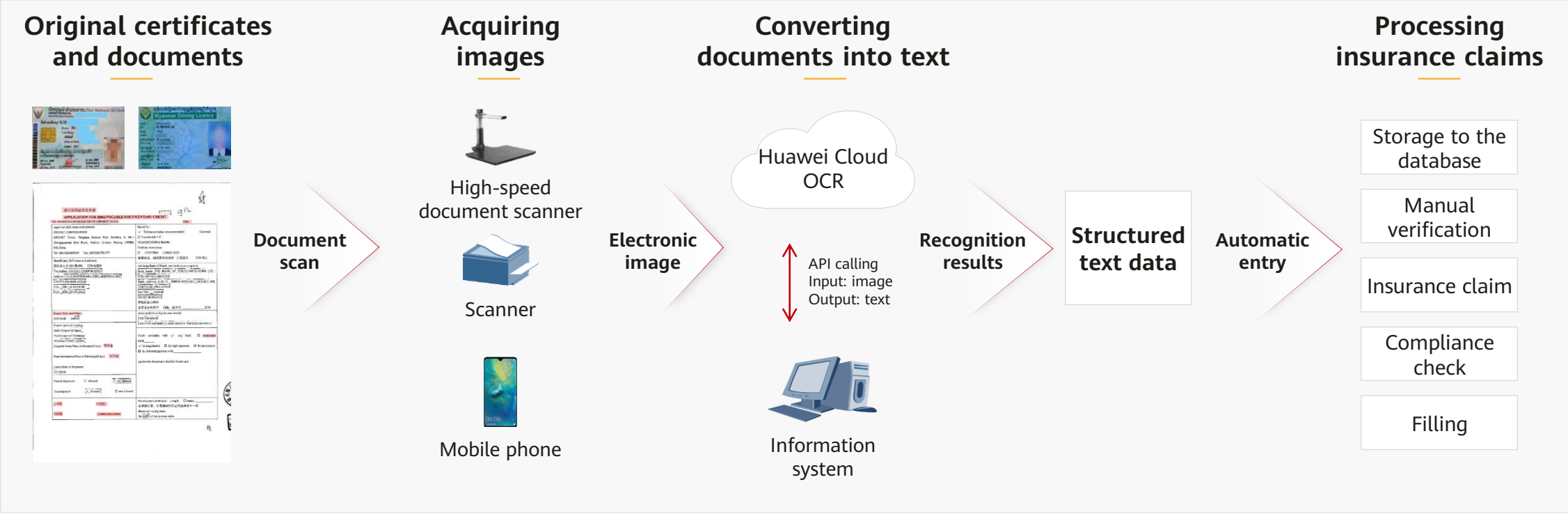
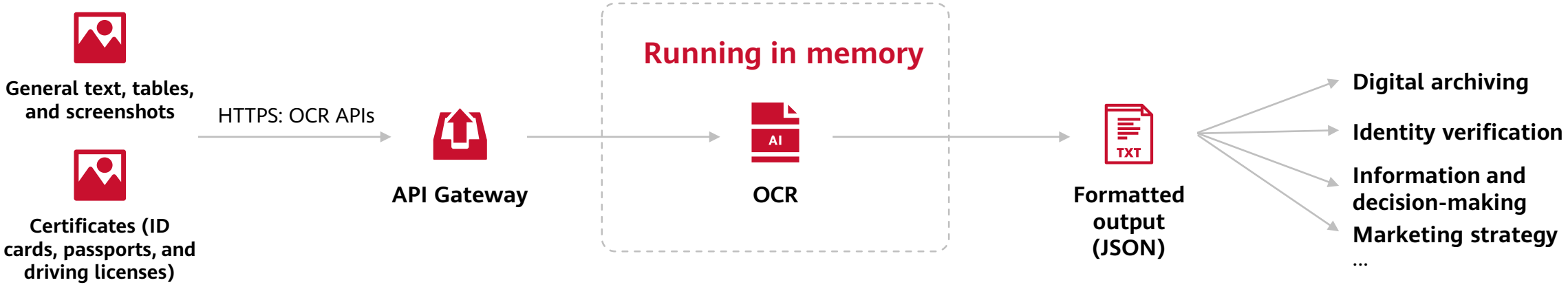
Introduction to the OCR Solution

Higher efficiency at lower costs

1. Many traditional industries, such as carriers, banks, and retail, have millions of scanned documents that need to be processed electronically for archiving. However, manual data entry is time-consuming and inefficient.
2. Real-time registration and review processes often involve multiple items that require manual entry, which can be costly and lead to a poor user experience.
3. In certain countries, ID cards may have handwritten information and poor-quality images, making them difficult to identify.



Introduction to the OCR Lite Solution



✓ **High accuracy**

Our advanced deep learning algorithms optimize various service scenarios to achieve highly accurate character recognition.

✓ **Support for complex scenarios**

Our OCR technology can recognize certificates with complex backgrounds, distortions, and tilts, as well as forms with seals and misaligned text.

✓ **Ease of use**

Our RESTful APIs are highly compatible and user-friendly.

✓ **Data security**

We only use customer data for identification purposes.

API call guidelines: https://support.huaweicloud.com/intl/en-us/api-ocr/ocr_03_0050.html

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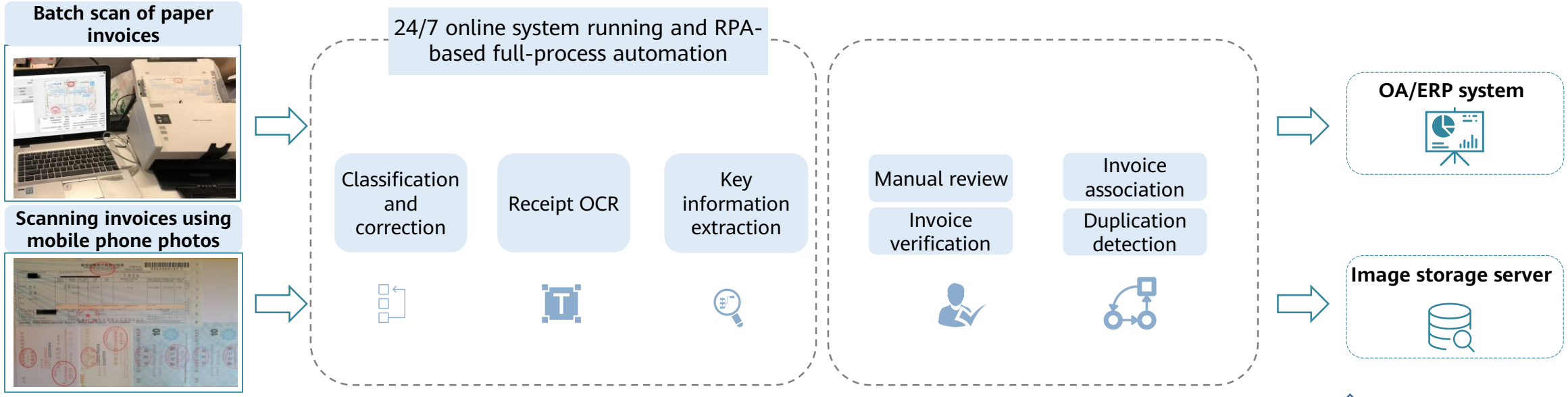
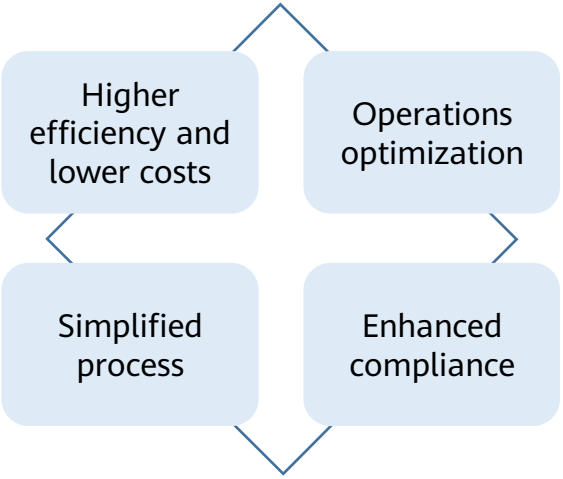
Use Case — Invoice OCR



- **Multiple access modes**: Connects to scanners, high-speed document scanners, or mobile phones for batch image acquisition.
- **Invoice variety**: Supports various invoice types, including VAT general/special/electronic invoices, ETC invoices, roll invoices, taxi invoices, train tickets, flight itineraries, quota invoices, and toll invoices.
- **Multiple invoices in one image**: Automatically identifies and classifies different invoice types that are combined in a single image.
- **Visual comparison**: Converts location data into an Excel file for easy statistical analysis.



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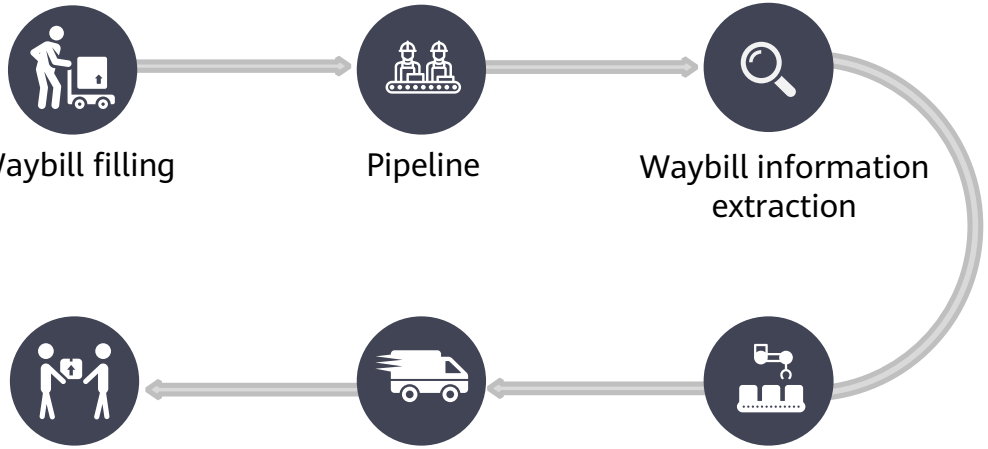
Use Case — Smart Logistics OCR

➢ID Card OCR

- ID card photographing, recognition, and verification with a mobile app

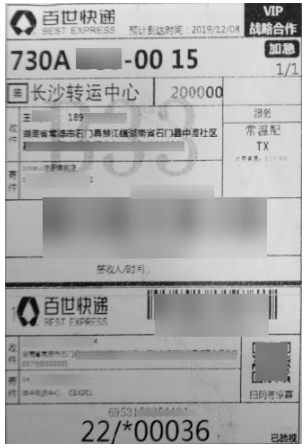


➢E-waybill OCR

- Automatically extract main information such as the waybill number, recipient/sender name, phone number, and address.
- Supports electronic waybills of mainstream courier companies.
- Recognizes multiple waybills concurrently.

➢Screenshot OCR

- Self-service parcel posting
- E-commerce sellers use this service to identify buyers' screenshots and automatically retrieve information such as the name and address.



24/7, less than 2s per waybill recognition

Up to 98% in accuracy, reducing the reshooting workload and eliminating external interference

Streamlined automation process, minimizing human intervention to reduce costs

Fully automated process, protecting data privacy

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Practice Case — Medical Insurance Claim Settlement

Original documents

ID card and bank card

Medical invoice

Other invoices

Document scan

Acquiring images

High-speed document scanner
Scanner
Mobile phone
...

Electronic image

Converting documents into text

Huawei Cloud OCR

API calling
Input: image
Output: text

Information system

Recognition results

Structured text data

Automatic entry

Processing insurance claims

- Storage to the database
- Manual verification
- Insurance claim
- Compliance check
- Filling

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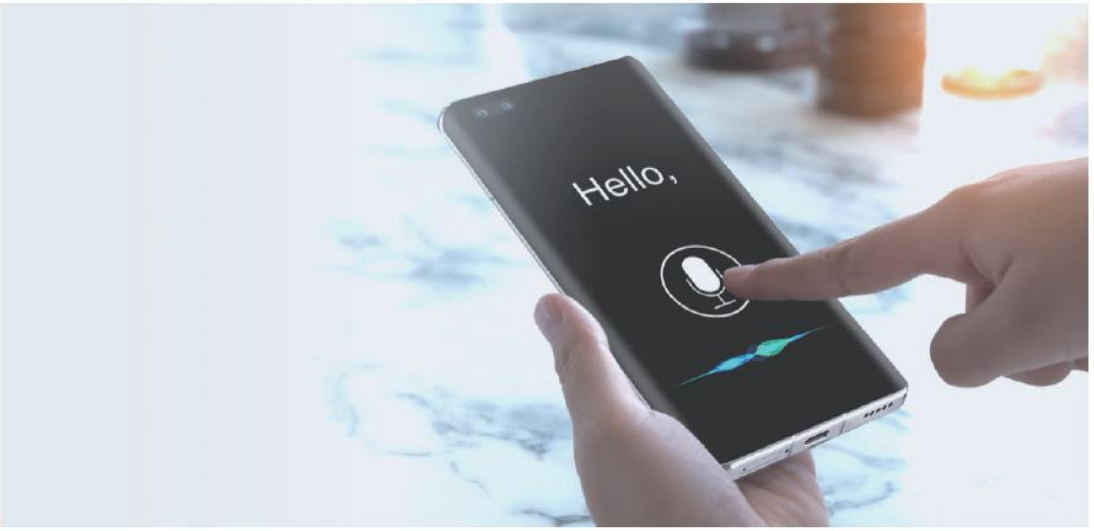
What Is SIS?

- SIS provides a man-machine interaction mode for users to obtain the speech interaction results through real-time access or API calling. For example, users can use Automatic Speech Recognition (ASR) to convert speech recordings or audio files into editable text, and use Text To Speech (TTS) to convert text into lifelike voices.



Intelligent algorithm

Traditional algorithms and deep learning are integrated for high-precision semantic match algorithms, ensuring high model accuracy.

Self-service optimization

Allows users to import hot words to optimize the recognition performance in specific domains.

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SIS Introduction

Function Module	Description	Application Scenario	Dialect
Real-time transcription	• Multiple modes: continuous sentences, streaming one-sentence, and single-sentence • Interruption tolerance • Silence detection • Intelligent text segmentation • Hot words customization	• Intelligent outbound call, real-time inspection, and conference transcription	• Sichuanese, Cantonese, and Shanghai dialect
Short sentence recognition	• Hot words customization • Vertical domain model customization	• Voice search, voice command, and voice SMS	• Sichuanese, Cantonese, and Shanghai dialect
Audio file transcription	• Emotion recognition • Speech speed recognition • Hot words customization • Silence detection • Speaker recognition	• Offline inspection and recording	• Sichuanese, Cantonese, and Shanghai dialect
Text to speech (TTS)	• Multiple timbres and customizable speech speed and volume	• Intelligent outbound call, audio reading, man-machine interaction, and voice navigation	• Sichuanese, Taiwan accent, and Southern Fujian dialect

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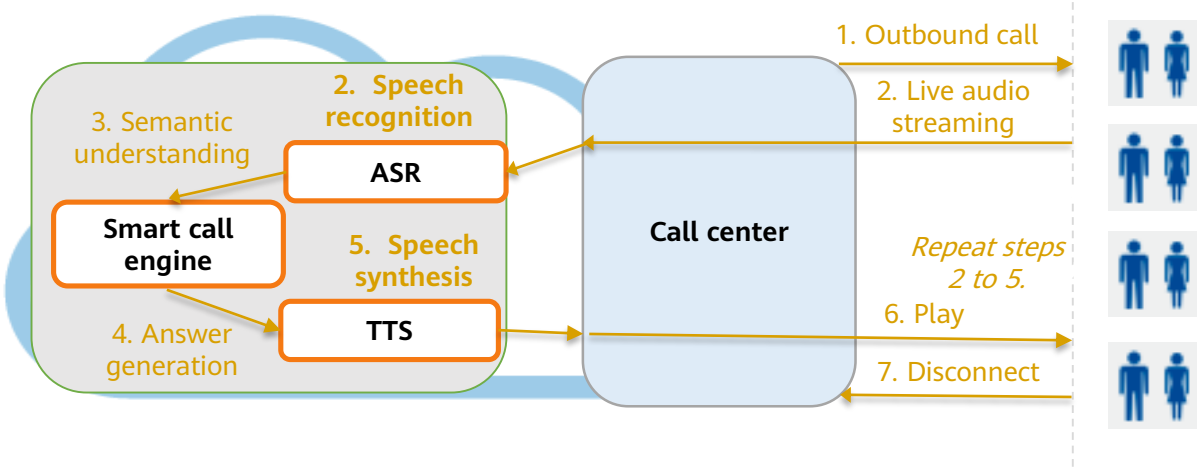
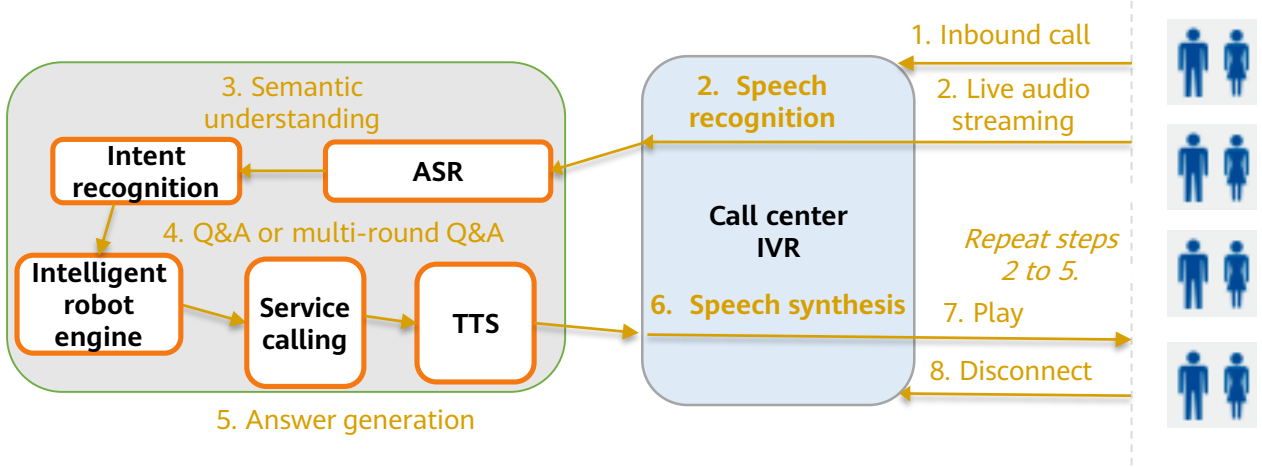
Application Scenario – Outgoing Calls and Incoming Calls

Architecture and process of an outbound call robot





Features

ASR and TTS

Advantages

Recognition rate: The dedicated outbound call model has been verified on a large scale, and the recognition rate is high.
Hot words: Hot words can be used to improve the accuracy of terminology recognition.
High concurrency and low latency: Elastic scaling of backend services meets high concurrency requirements. Real-time identification features low latency, high concurrency, and response within milliseconds.
Dedicated timbres: Up to 16 synthesis timbres are available for the customer service.

Customer benefits

Reduced cost
Improved efficiency: Robots' 24/7 response within milliseconds
Increased service satisfaction: Customer complaint rate decreased by 30%

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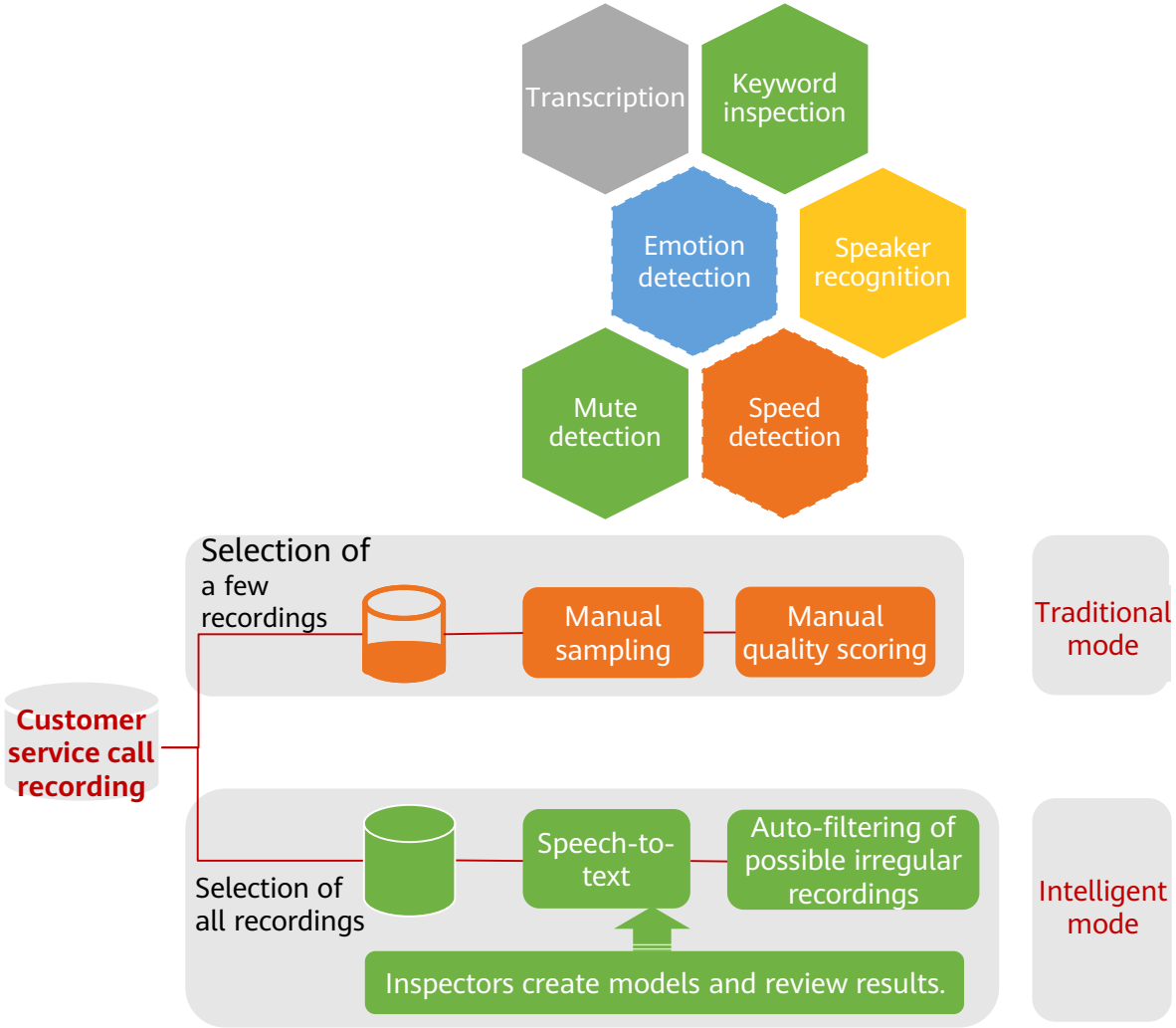
Application Scenario – Customer Service Inspection



Features

Audio file transcription

Advantages

Speaker separation: 8K speaker recognition model that can automatically identify the voice of customers and the customer service.
Emotion detection
High recognition rate: The vertical-domain dedicated model has been verified on a large scale, and the recognition rate is high.

Customer benefits

Reduced cost
Improve efficiency: All recordings are checked to avoid omissions.

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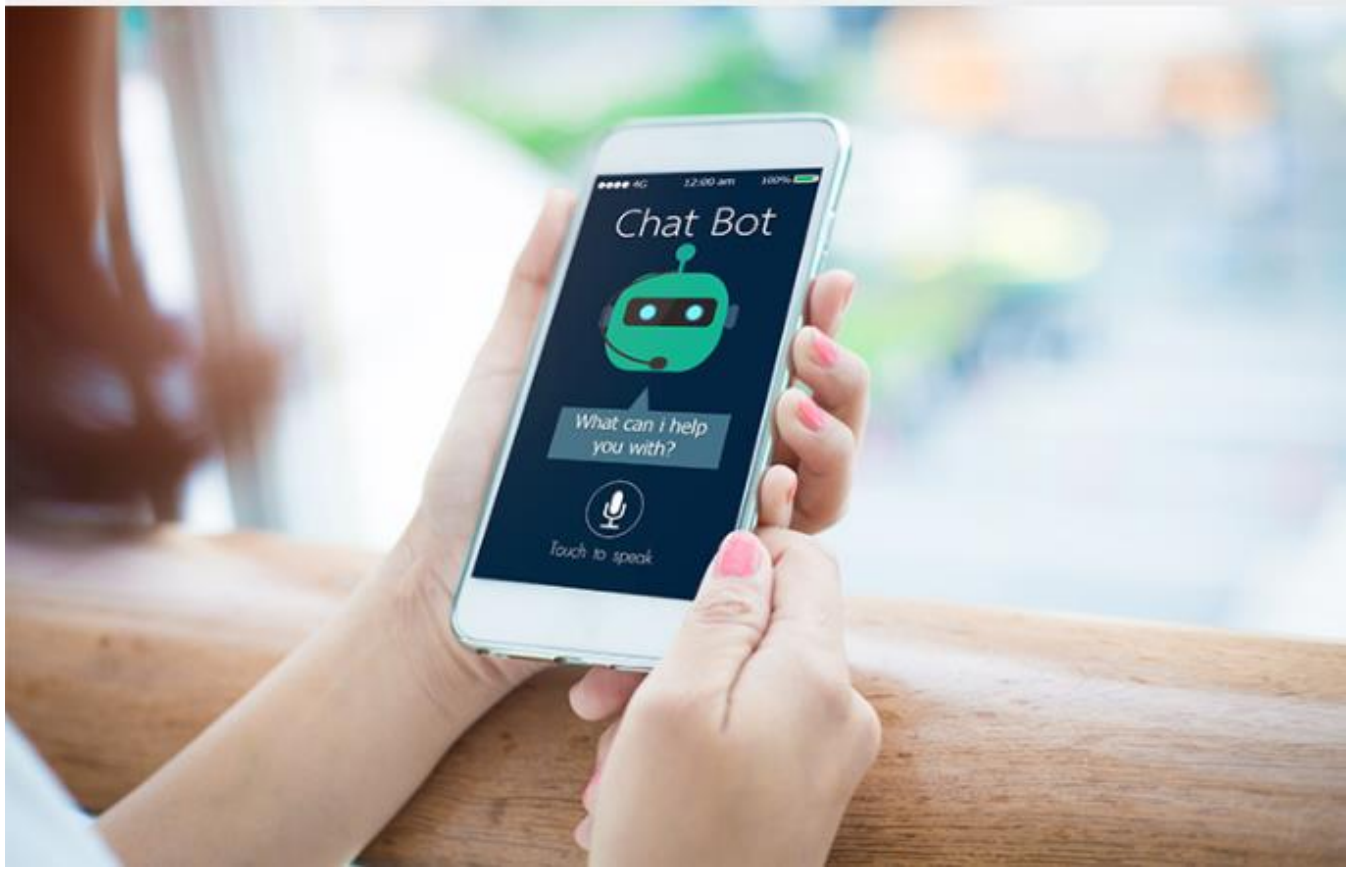
Application Scenario – Man-Machine Interaction



Voice search, voice command, and voice SMS

Features

Short sentence recognition, real-time speech recognition, and speech synthesis

Advantages

VAD: The real-time single-sentence recognition has the voice activity detection (VAD) function, which can identify the start and end of a sentence.
Easy access: WebSocket and HTTP access are supported to meet requirements in different scenarios.
Diverse audio formats: Supports 11 audio formats, such as MP3 and WAV.
High recognition rate: The dedicated man-machine interaction model makes recognition more accurate.

Customer benefits

Extended input: The voice recognition function is integrated.

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Application Scenario – Audio Reading

Alvin male voice

Ahmed male voice

Amy female voice

Aisha female voice

Features

Text to speech (TTS)

Advantages

Cloud-based synthesis: The voice synthesized on cloud can be used at anytime.
Exclusive timbre: Exclusive timbre and lifelike voices for audio reading

Customer benefits

Shorten production time: Supports quick production of audio reading works.

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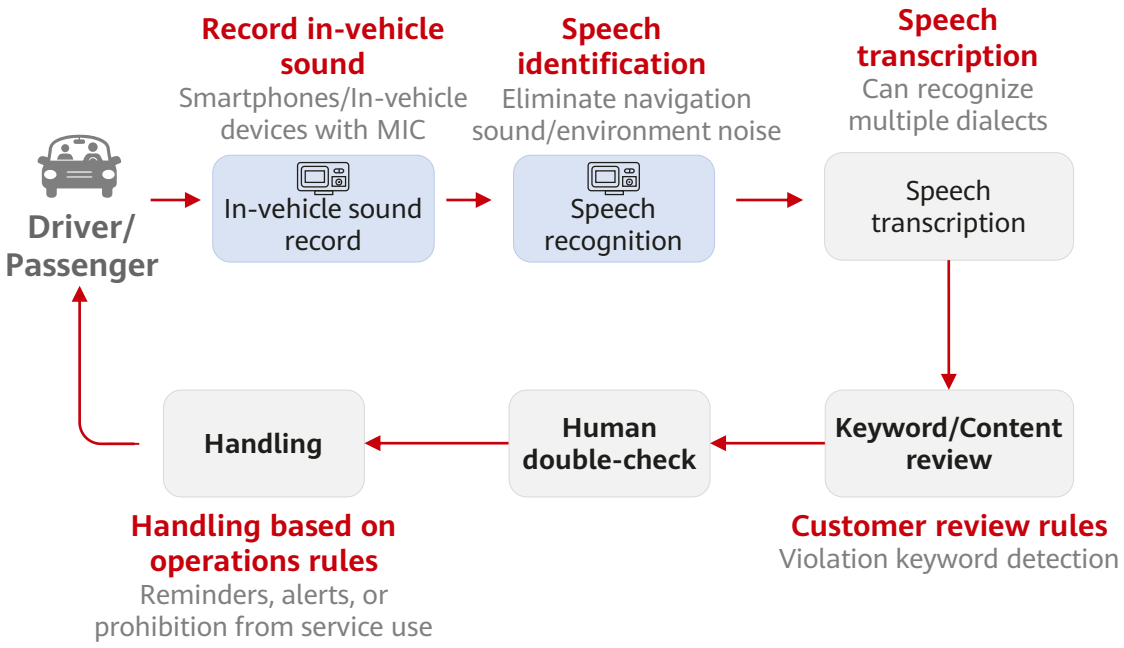
Practice Case – Top Ride-Hailing Platform

Background

- **Requirements**: accurate identification of sensitive words or violations.
- **Pain points**: noise in the audio files recorded by mobile phones or in-vehicle devices; low accuracy of real-time/quasi-real-time speech transcription

Solution Architecture



Solution highlights

- **High accuracy**: up to 89% accuracy with a 2-hour test set in complex scenarios
- **Low cost**: filtering models are configured on devices for higher efficiency
- **Cost-effectiveness**: diverse commercial discounts

Customer benefits

- Voice inspection solutions with high accuracy in noisy environments

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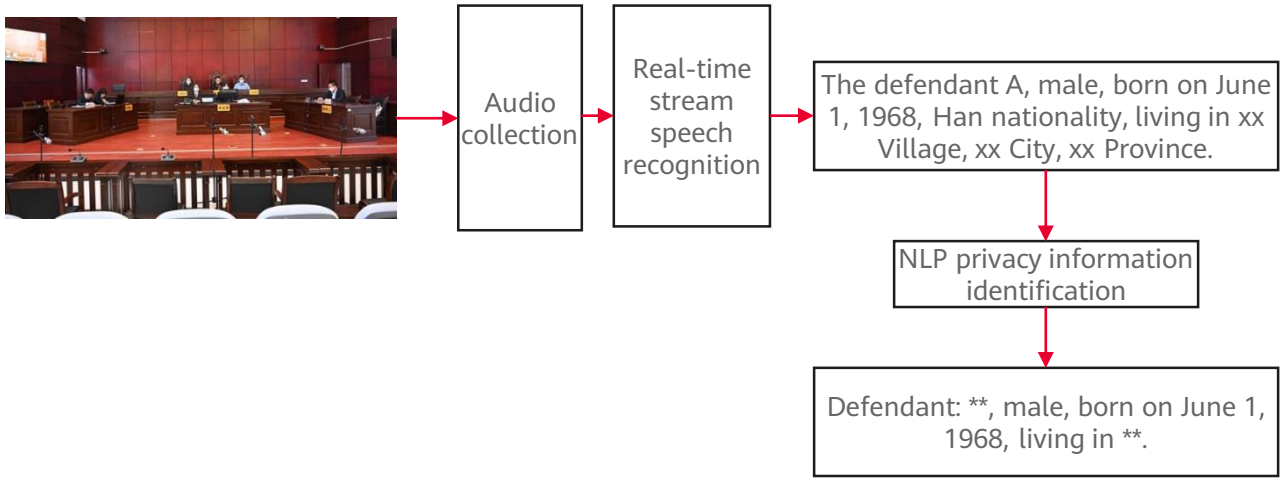
Practice Case – Top Trial Live Stream Platform

Background

- **Requirements**: real-time speech recognition, NLP privacy information identification
- **Pain points**: high concurrency requirements

Solution Architecture



Solution highlights

- **High accuracy**: industry-leading speech recognition accuracy in the case of court hearings
- **Customizable**: The NLP algorithm for privacy information anonymization can be customized based on service requirements.
- **High concurrency**: Supported 5000+ trial live streaming in China, implementing elastic scaling and on-demand charging.

Customer benefits

- Subtitle anonymization solution for trial live streams that supports high concurrency

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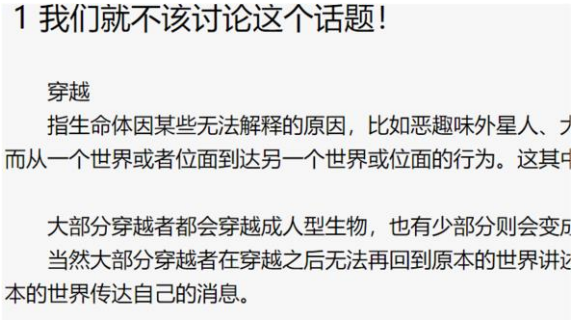
Practice Case – Audiobook Application


1 我们就不该讨论这个话题！

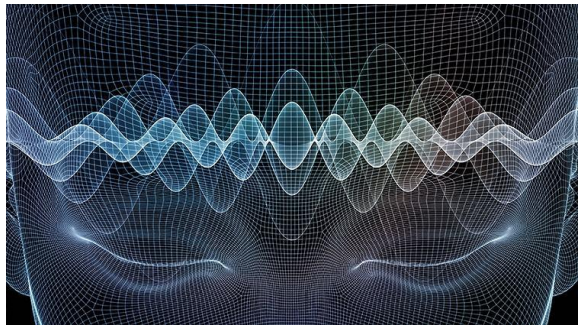
穿越

指生命体因某些无法解释的原因，比如恶趣味外星人、大
而从一个世界或者位面到达另一个世界或位面的行为。这其中

大部分穿越者都会穿越成人型生物，也有少部分则会变成
当然大部分穿越者在穿越之后无法再回到原本的世界讲这
本的世界传达自己的消息。

E-book text

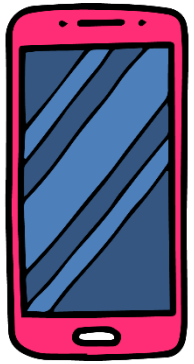
RESTful API



Text-to-Speech

Audio file

Cloud storage



Download when reading

1. Synthesize once and use anytime.
2. Synthesize smooth and natural timbre on cloud.

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Contents

1. Image Recognition
2. ImageSearch
3. Content Moderation
4. OCR
5. SIS
6. **Invoking API Services**

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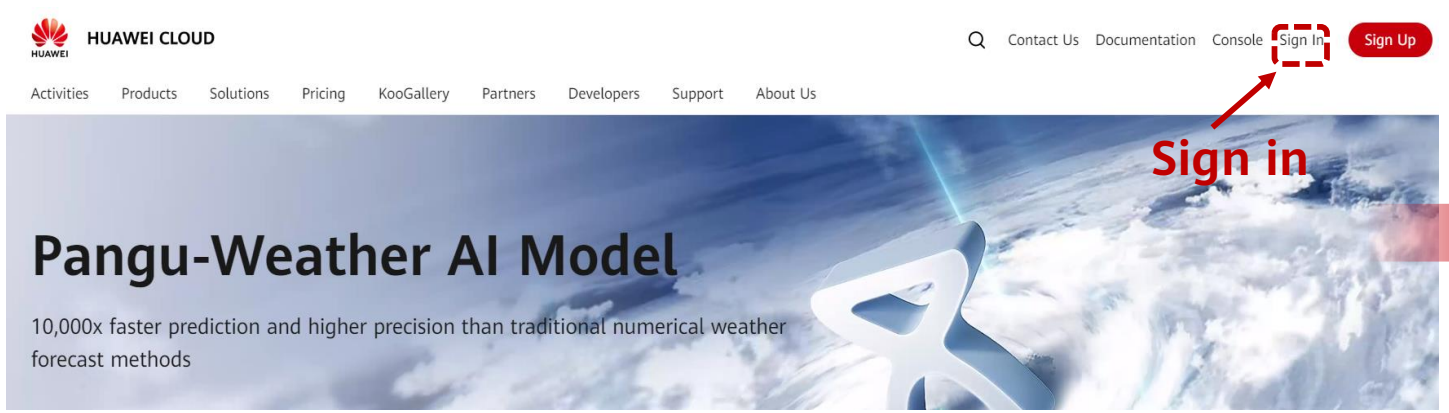
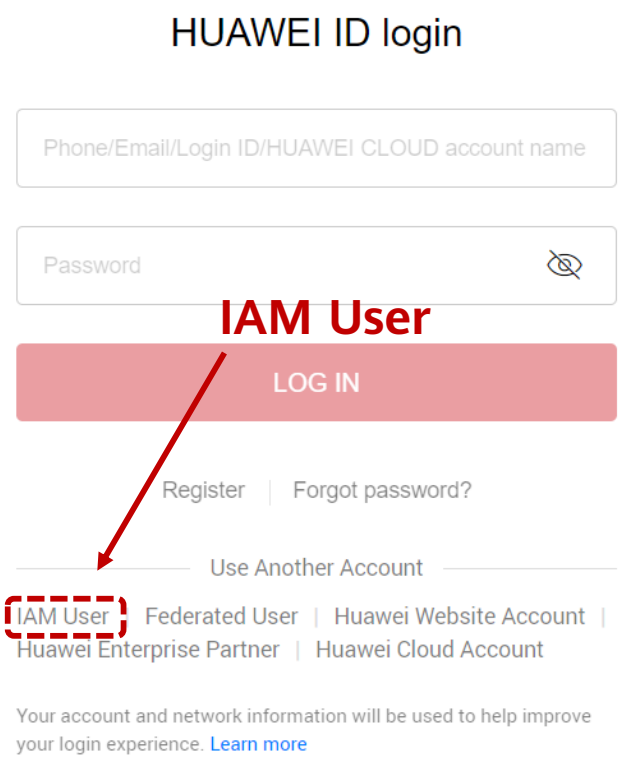
API Calling Method

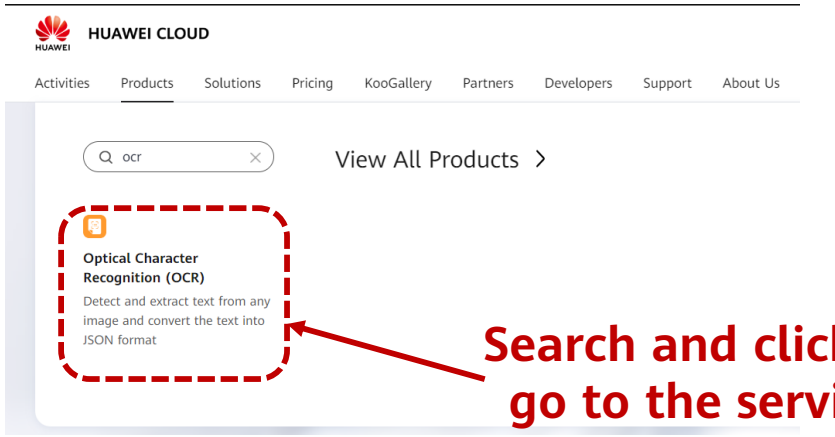
Huawei Cloud official website
https://www.huaweicloud.com/intl/en-us/





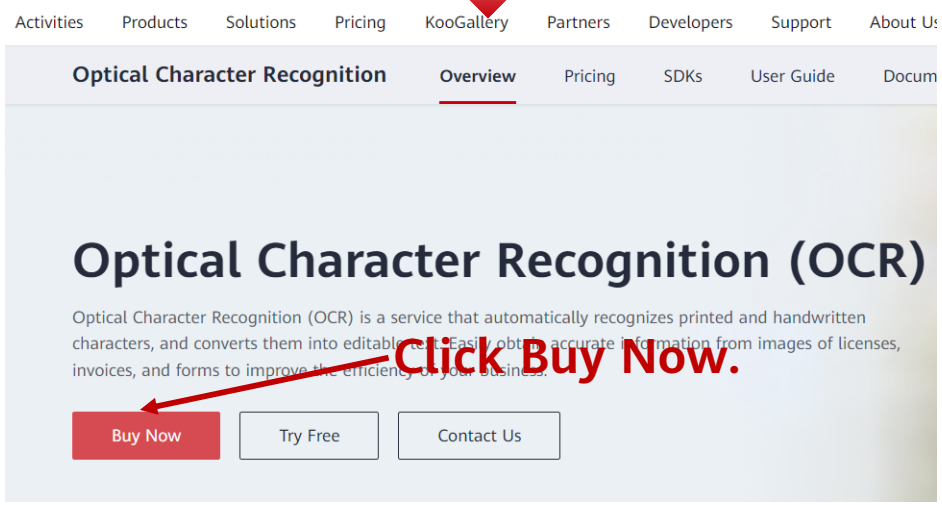
API Calling Method (2) — Using OCR as an Example

Sign In → **Subscribe** → Call



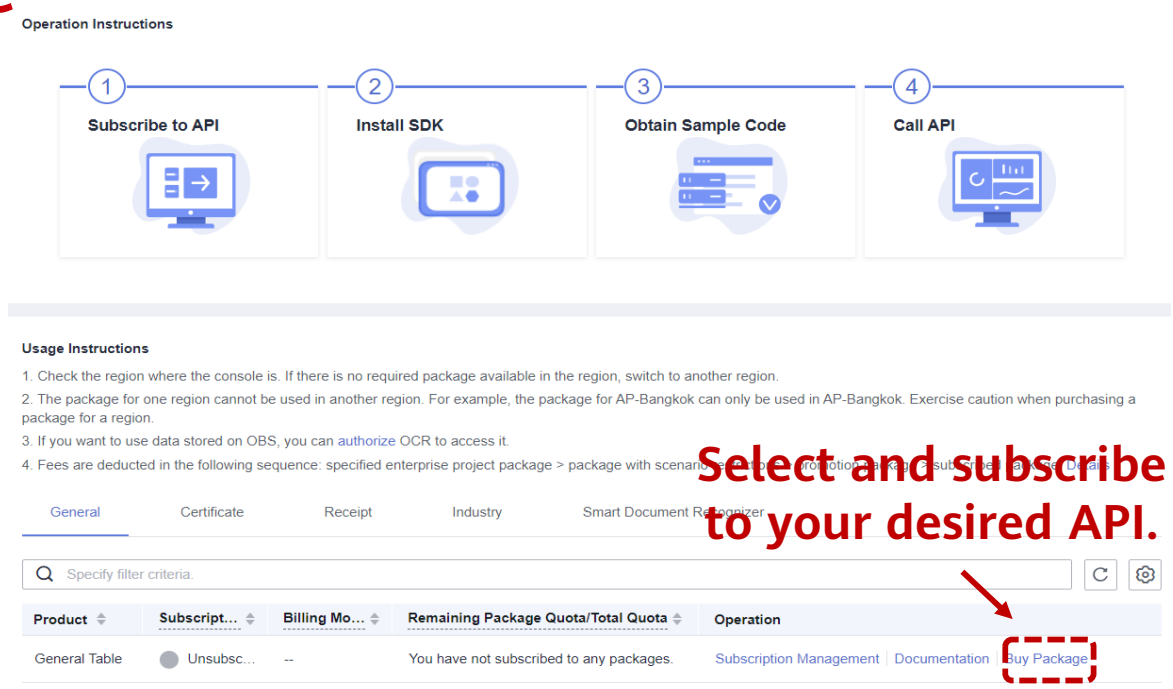
Search and click to go to the service details page.

1



Click Buy Now.

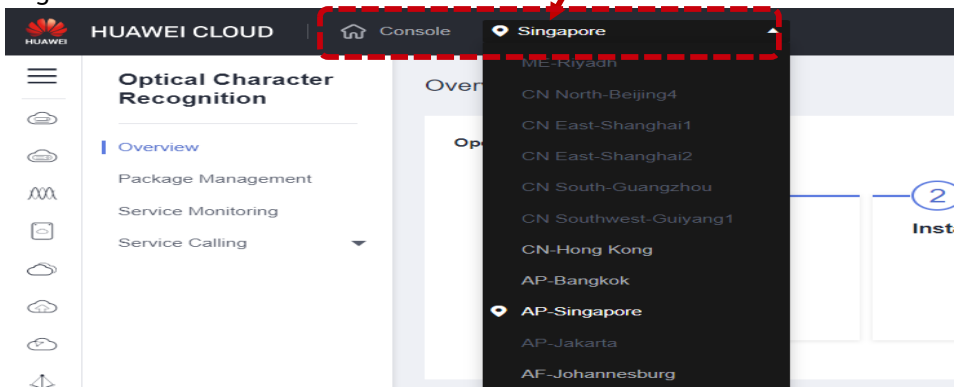
2



Select and subscribe to your desired API.

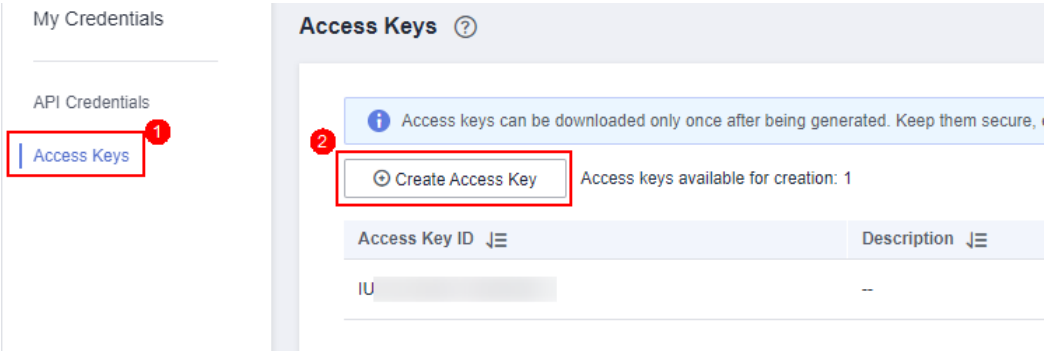
1. Check the region where the console is. If there is no required package available in the region, switch to another region.
2. The package for one region cannot be used in another region. For example, the package for AP-Bangkok can only be used in AP-Bangkok. Exercise caution when purchasing a package for a region.

3



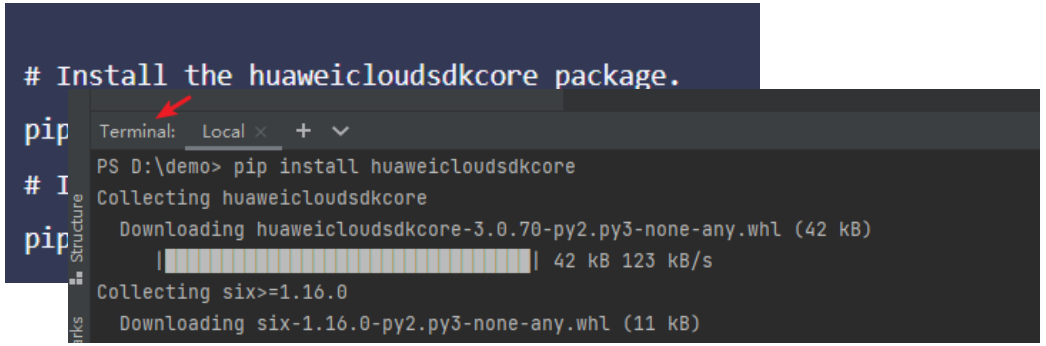
API Calling Method (3) — Using OCR as an Example

Obtain an AK and SK on the **My Credentials** > **Access Keys** page.



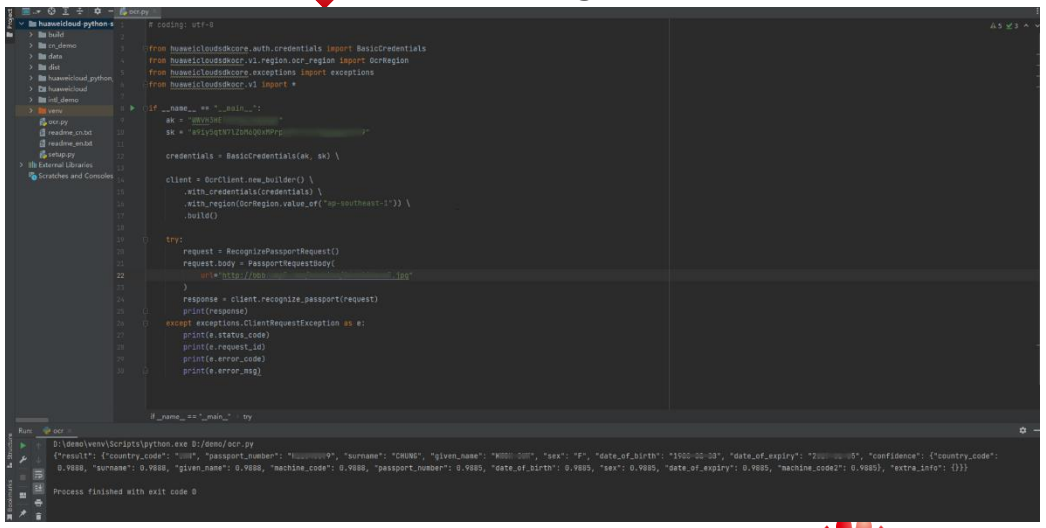
A	B	C	D	E
User Name	Access Ke	Secret Access Key		
testuser	LSKM	rIZaQ		
	AK	SK		

Install the SDK.



1

2 Run the sample code to obtain recognition results.



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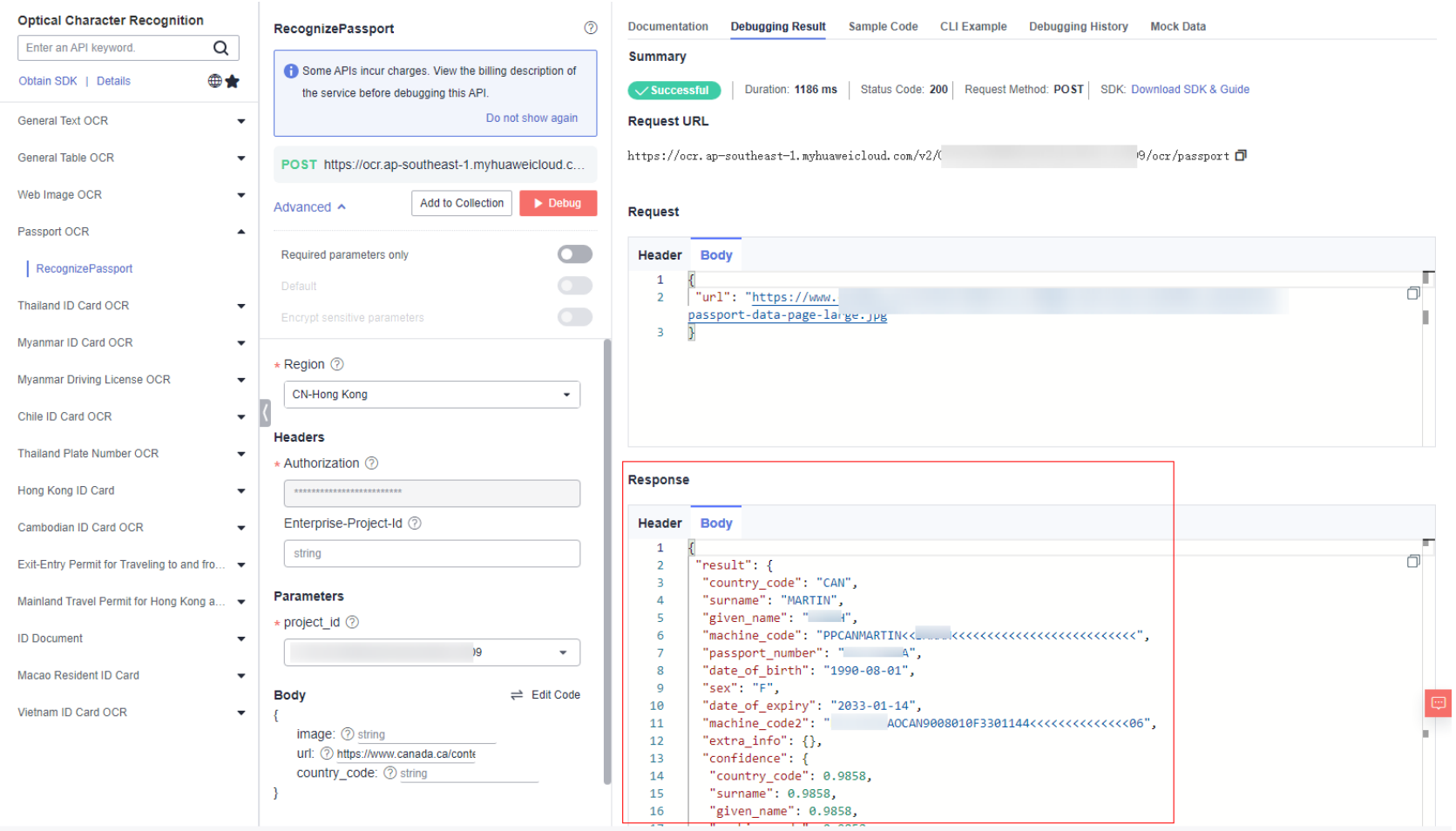
Huawei Cloud API Explorer

API Explorer: an online debugging tool for retrieving and debugging APIs and generating example code
https://console-intl.huaweicloud.com/apiexplorer/#/openapi



Q&A

1. Which of the following API services are provided by Huawei Cloud? ( )

   A. Speech Interaction Service (SIS)

   B. Content Moderation

   C. Optical Character Recognition (OCR)

   D. ImageSearch

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Q&A

2. What are the application scenarios of Huawei Cloud Image Recognition?

   A. Scenario analysis

   B. Image tagging

   C. Object detection

   D. Smart album

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Q&A

1. What are the benefits of using an OCR-based system for financial reimbursements? ( )

   A. Multiple tickets or certificates in a single image

   B. Support for automatic segmentation

   C. Multiple access modes

   D. Support for integration with an OA system and a storage server

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Summary

- This chapter describes the functions and application scenarios of Huawei Cloud API services, including Image Recognition, ImageSearch, Content Moderation, OCR, and SIS. This chapter also describes how to subscribe to an API service on Huawei Cloud and call an API.

Acronyms and Abbreviations

- EI: enterprise intelligence
- API: application programming interface
- GCN: graph convolutional network
- IM: instant message
- RPA: robotic process automation
- OCR: optical character recognition

Recommendations

- Image Recognition documentation:
  - https://support.huaweicloud.com/intl/en-us/productdesc-image/image_01_0002.html
- ImageSearch documentation:
  - https://support.huaweicloud.com/intl/en-us/imagesearch/index.html
- Content Moderation documentation:
  - https://support.huaweicloud.com/intl/en-us/moderation/index.html
- OCR documentation:
  - https://support.huaweicloud.com/intl/en-us/ocr/index.html
- SIS documentation:
  - https://support.huaweicloud.com/intl/en-us/sis/index.html

Thank You.

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